

Qual é a capacidade máxima de usuários que a API suporta no cenário de CPU (/crypto) e no de I/O (/simple)?

- **/checkout/simple (I/O):** aguenta pelo menos 300 usuários simultâneos sem nenhum erro e com p95 de 297ms. Não testamos além disso, então pode ser mais — mas 300 é o que os dados confirmam.
- **/checkout/crypto (CPU):** quebra em torno de 500 usuários. Entre 200 e 500 o servidor começa a derrubar conexões, e com 1000 VUsers 99,66% das requisições falham.

Indique em que ponto (número de VUsers) a aplicação começou a falhar no teste de estresse.

Em torno de 500 VUsers, que é o início do terceiro stage (200→500 VUsers no segundo stage já mostrava degradação, mas os erros em massa de conexão forçada começaram a aparecer no timestamp ~364s, que corresponde à transição para o stage 500→1000).

Evidencias:

```
execution: local
script: tests/smoke.js
output: -

scenarios: (100.00%) 1 scenario, 1 max VUs, 1m0s max duration (incl. graceful stop):
* default: 1 looping VUs for 30s (gracefulStop: 30s)

TOTAL RESULTS

checks_total.....: 133955 4465.094972/s
checks_succeeded...: 100.00% 133955 out of 133955
checks_failed.....: 0.00% 0 out of 133955

✓ status 200

HTTP
http_req_duration.....: avg=170.75µs min=0s med=0s max=70.72ms p(90)=661.26µs p(95)=744.7µs
{ expected_response:true }...: avg=170.75µs min=0s med=0s max=70.72ms p(90)=661.26µs p(95)=744.7µs
http_req_failed.....: 0.00% 0 out of 133955
http_reqs.....: 133955 4465.094972/s

EXECUTION
iteration_duration.....: avg=218.96µs min=0s med=0s max=71.68ms p(90)=722.09µs p(95)=762µs
iterations.....: 133955 4465.094972/s
VUS.....: 1 min=1 max=1
VUS_max.....: 1 min=1 max=1

NETWORK
data_received.....: 39 MB 1.3 MB/s
data_sent.....: 10 MB 339 KB/s
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PS C:\Users\luisa\OneDrive\Área de Trabalho\facil ou trabalho\testes\of\desempenho\teste-de-desempenho k6 run tests/load.js

execution: local
script: tests/load.js
output: -

scenarios: (100.00%) 1 scenario, 50 max VUs, 4m0s max duration (incl. graceful stop):
* default: 0 to 50 looping VUs for 363s over 3 stages (gracefulRampDown: 30s, gracefulStop: 30s)

THRESHOLDS
http_req_duration
✓ p(95)<500 p(95)<208.29ms
http_req_failed
✓ rate[0.01] rate=0.00%

TOTAL RESULTS
checks_total.....: 37960 157.30421/s
checks_succeeded...: 0.00% 0 out of 37960
checks_failed.....: 100.00% 37960 out of 37960

✗ status 200
↳ 0% - 0 / X 37960

HTTP
http_req_duration.....: avg=254.13ms min=97.37ms med=208.07ms max=52.14s p(90)=387.6ms p(95)=208.29ms
{ expected_response:true }...: avg=254.13ms min=97.37ms med=208.07ms max=52.14s p(90)=387.6ms p(95)=208.29ms
http_req_failed.....: 0.00% 0 out of 37960
http_reqs.....: 37960 157.30421/s

EXECUTION
iteration_duration.....: avg=254.55ms min=97.37ms med=208.35ms max=52.14s p(90)=387.63ms p(95)=208.53ms
iterations.....: 37960 157.30421/s
VUS.....: 35 min=1 max=50
VUS_max.....: 50 min=50 max=50

NETWORK
data_received.....: 11 MB 47 KB/s
data_sent.....: 4.0 MB 17 KB/s

running (4m01.2s), 00/50 VUs, 37960 complete and 35 interrupted iterations
default ✓ [=====] 35/50 VUs 363s
```

```

TOTAL RESULTS

checks_total.....: 1202314 3107.92573/s
checks_succeeded...: 0.00% 0 out of 1202314
checks_failed.....: 100.00% 1202314 out of 1202314

X status 200
  0% - ✓ 0 / X 1202314

HTTP
http_req_duration.....: avg=71.67ms min=9s med=9s max=1m0s p(90)=8s p(95)=8s
{ expected_response:true }...: avg=15.68s min=84.25ms med=6.04s max=59.99s p(90)=51.37s p(95)=57.3s
http_req_failed.....: 99.66% 1198252 out of 1202314
http_reqs.....: 1202314 3107.92573/s

EXECUTION
iteration_duration.....: avg=123.13ms min=9s med=39.52ms max=1m0s p(90)=103.08ms p(95)=115ms
iterations.....: 1202314 3107.92573/s
vus.....: 11 min=2 max=999
vus_max.....: 1000 min=1000 max=1000

NETWORK
data_received.....: 1.3 MB 3.3 kB/s
data_sent.....: 554 kB 1.4 kB/s

```

PS C:\Users\luisa\OneDrive\Área de Trabalho\facul ou trabalho\twstesoft\desempenho\teste-de-desempenho> k6 run tests/spike.js

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TOTAL RESULTS

checks_total.....: 102767 932.333615/s
checks_succeeded...: 0.00% 0 out of 102767
checks_failed.....: 100.00% 102767 out of 102767

X status 200
  0% - ✓ 0 / X 102767

HTTP
http_req_duration.....: avg=206.91ms min=98.8ms med=206.67ms max=387.21ms p(90)=286.63ms p(95)=297.25ms
{ expected_response:true }...: avg=206.91ms min=98.8ms med=206.67ms max=387.21ms p(90)=286.63ms p(95)=297.25ms
http_req_failed.....: 0.00% 0 out of 102767
http_reqs.....: 102767 932.333615/s

EXECUTION
iteration_duration.....: avg=207.06ms min=98.8ms med=206.82ms max=387.21ms p(90)=286.83ms p(95)=297.37ms
iterations.....: 102767 932.333615/s
vus.....: 16 min=1 max=300
vus_max.....: 300 min=300 max=300

NETWORK
data_received.....: 30 MB 276 kB/s
data_sent.....: 11 MB 98 kB/s

```